

Movie_Genre_Classification - Co

Ask Gemini

colab.research.google.com/drive/1--YgBCA3dCppMQjcxP5J4tD2-RRYqolQ#scrollTo=sMiqk93kjP_P

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CO

Movie_Genre_Classification

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✓ RAM Disk

Files

📁 ..

📁 sample_data

📄 movie_genre_model.pkl

📄 prediction_results.csv

📄 tfidf_vectorizer.pkl

📄 train_data.txt

Disk 87.25 GB available

Dataset Loaded Successfully

	ID	TITLE	GENRE
0	1	Oscar et la dame rose (2009)	drama
1	2	Cupid (1997)	thriller
2	3	Young, Wild and Wonderful (1980)	adult
3	4	The Secret Sin (1915)	drama
4	5	The Unrecovered (2007)	drama

DESCRIPTION

0

Listening in to a conversation between his doc...

1

A brother and sister with a past incestuous re...

2

As the bus empties the students for their fiel...

3

To help their unemployed father make ends meet...

4

The film's title refers not only to the un-rec...

Dataset Shape: (54214, 4)

Missing Values:

ID

0

TITLE

0

GENRE

0

DESCRIPTION

0

dtype: int64

Genre Count:

GENRE

drama

13613

documentary

13096

comedy

7447

short

5073

horror

2204

thriller

1591

Variables

Terminal

11:10 Python 3

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documentary 13096

comedy 7447

short 5073

horror 2204

thriller 1591

action 1315

western 1032

reality-tv 884

family 784

adventure 775

music 731

romance 672

sci-fi 647

adult 590

crime 505

animation 498

sport 432

talk-show 391

fantasy 323

mystery 319

musical 277

biography 265

history 243

game-show 194

news 181

war 132

Name: count, dtype: int64

Cleaned Text Example:

DESCRIPTION \

0 Listening in to a conversation between his doc...

1 A brother and sister with a past incestuous re...

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{ } Variables

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```
DESCRIPTION
0 Listening in to a conversation between his doc...
1 A brother and sister with a past incestuous re...
2 As the bus empties the students for their fiel...
3 To help their unemployed father make ends meet...
4 The film's title refers not only to the un-rec...

CLEAN_DESCRIPTION
0 listening in to a conversation between his doc...
1 a brother and sister with a past incestuous re...
2 as the bus empties the students for their fiel...
3 to help their unemployed father make ends meet...
4 the films title refers not only to the unreco...

Training Samples: 43371
Testing Samples: 10843

TF-IDF Feature Shape: (43371, 70000)

=====
Model: Naive Bayes
=====
Accuracy: 0.4593747117956285

Classification Report:
              precision    recall  f1-score   support

   action           0.00      0.00      0.00        263
    adult           0.00      0.00      0.00        118
  adventure         0.00      0.00      0.00        155
  animation         0.00      0.00      0.00        100
  biography         0.00      0.00      0.00         53
  comedy           0.00      0.00      0.00        100
```

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Classification Report:

	precision	recall	f1-score	support
action	0.00	0.00	0.00	263
adult	0.00	0.00	0.00	118
adventure	0.00	0.00	0.00	155
animation	0.00	0.00	0.00	100
biography	0.00	0.00	0.00	53
comedy	0.66	0.12	0.21	1490
crime	0.00	0.00	0.00	101
documentary	0.53	0.92	0.67	2619
drama	0.40	0.87	0.55	2723
family	0.00	0.00	0.00	157
fantasy	0.00	0.00	0.00	65
game-show	0.00	0.00	0.00	39
history	0.00	0.00	0.00	49
horror	1.00	0.00	0.00	441
music	1.00	0.01	0.03	146
musical	0.00	0.00	0.00	55
mystery	0.00	0.00	0.00	64
news	0.00	0.00	0.00	36
reality-tv	0.00	0.00	0.00	177
romance	0.00	0.00	0.00	134
sci-fi	0.00	0.00	0.00	129
short	0.91	0.01	0.02	1015
sport	0.67	0.02	0.04	86
talk-show	0.00	0.00	0.00	78
thriller	0.00	0.00	0.00	318
war	0.00	0.00	0.00	26
western	1.00	0.01	0.02	206
accuracy	0.16	0.16	0.16	10813

Variables

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sport	0.67	0.02	0.04	86
talk-show	0.00	0.00	0.00	78
thriller	0.00	0.00	0.00	318
war	0.00	0.00	0.00	26
western	1.00	0.01	0.02	206
accuracy			0.46	10843
macro avg	0.23	0.07	0.06	10843
weighted avg	0.48	0.46	0.33	10843

=====
Model: Logistic Regression
=====
Accuracy: 0.5773309969565619

Classification Report:

	precision	recall	f1-score	support
action	0.60	0.22	0.32	263
adult	0.76	0.26	0.39	118
adventure	0.75	0.10	0.17	155
animation	1.00	0.02	0.04	100
biography	0.00	0.00	0.00	53
comedy	0.52	0.58	0.54	1490
crime	0.50	0.01	0.02	101
documentary	0.65	0.88	0.75	2619
drama	0.53	0.80	0.64	2723
family	0.71	0.08	0.14	157
fantasy	0.00	0.00	0.00	65
game-show	1.00	0.28	0.44	39
history	0.00	0.00	0.00	49

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news	0.00	0.00	0.00	36
reality-tv	0.55	0.13	0.21	177
romance	0.00	0.00	0.00	134
sci-fi	0.56	0.17	0.26	129
short	0.51	0.28	0.36	1015
sport	0.60	0.17	0.27	86
talk-show	0.86	0.08	0.14	78
thriller	0.40	0.10	0.16	318
war	0.00	0.00	0.00	26
western	0.95	0.60	0.73	206
accuracy			0.58	10843
macro avg	0.47	0.21	0.25	10843
weighted avg	0.56	0.58	0.53	10843

=====
Model: Support Vector Machine
=====
Accuracy: 0.5820344922991791

Classification Report:
precision recall f1-score support
action 0.48 0.37 0.42 263
adult 0.68 0.44 0.54 118
adventure 0.45 0.21 0.29 155
animation 0.49 0.17 0.25 100
biography 0.00 0.00 0.00 53
comedy 0.53 0.59 0.55 1490
crime 0.25 0.05 0.08 101
documentary 0.69 0.83 0.75 2619

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Commands Code Text Run all

RAM Disk

Files

sample_data movie_genre_model.pkl prediction_results.csv tfidf_vectorizer.pkl train_data.txt

Disk 87.25 GB available

fantasy	0.10	0.02	0.03	65
game-show	0.92	0.59	0.72	39
history	0.40	0.04	0.07	49
horror	0.61	0.64	0.62	441
music	0.63	0.50	0.56	146
musical	0.20	0.04	0.06	55
mystery	0.17	0.02	0.03	64
news	0.33	0.08	0.13	36
reality-tv	0.43	0.24	0.31	177
romance	0.23	0.05	0.09	134
sci-fi	0.49	0.36	0.42	129
short	0.44	0.34	0.38	1015
sport	0.59	0.45	0.51	86
talk-show	0.37	0.19	0.25	78
thriller	0.34	0.17	0.23	318
war	0.80	0.15	0.26	26
western	0.83	0.78	0.80	206
accuracy			0.58	10843
macro avg	0.46	0.30	0.34	10843
weighted avg	0.55	0.58	0.55	10843
Model: Random Forest				
Accuracy: 0.4911924744074518				
Classification Report:				
	precision	recall	f1-score	support
action	0.80	0.02	0.03	263

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sample_data

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drama	0.42	0.84	0.56	2723
family	1.00	0.02	0.04	157
fantasy	0.00	0.00	0.00	65
game-show	1.00	0.36	0.53	39
history	0.00	0.00	0.00	49
horror	0.73	0.15	0.25	441
music	0.69	0.08	0.14	146
musical	0.50	0.02	0.04	55
mystery	0.00	0.00	0.00	64
news	0.00	0.00	0.00	36
reality-tv	0.00	0.00	0.00	177
romance	0.00	0.00	0.00	134
sci-fi	0.50	0.02	0.04	129
short	0.65	0.09	0.16	1015
sport	0.61	0.13	0.21	86
talk-show	1.00	0.05	0.10	78
thriller	0.43	0.02	0.04	318
war	0.00	0.00	0.00	26
western	0.94	0.35	0.51	206
accuracy			0.49	10843
macro avg	0.47	0.13	0.15	10843
weighted avg	0.53	0.49	0.40	10843

Model Accuracy Comparison:
Naive Bayes : 0.4593747117956285
Logistic Regression : 0.5773309969565619
Support Vector Machine : 0.5820344922991791
Random Forest : 0.4911924744074518

Best Model: Support Vector Machine

Variables

Terminal

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Files

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sample_data

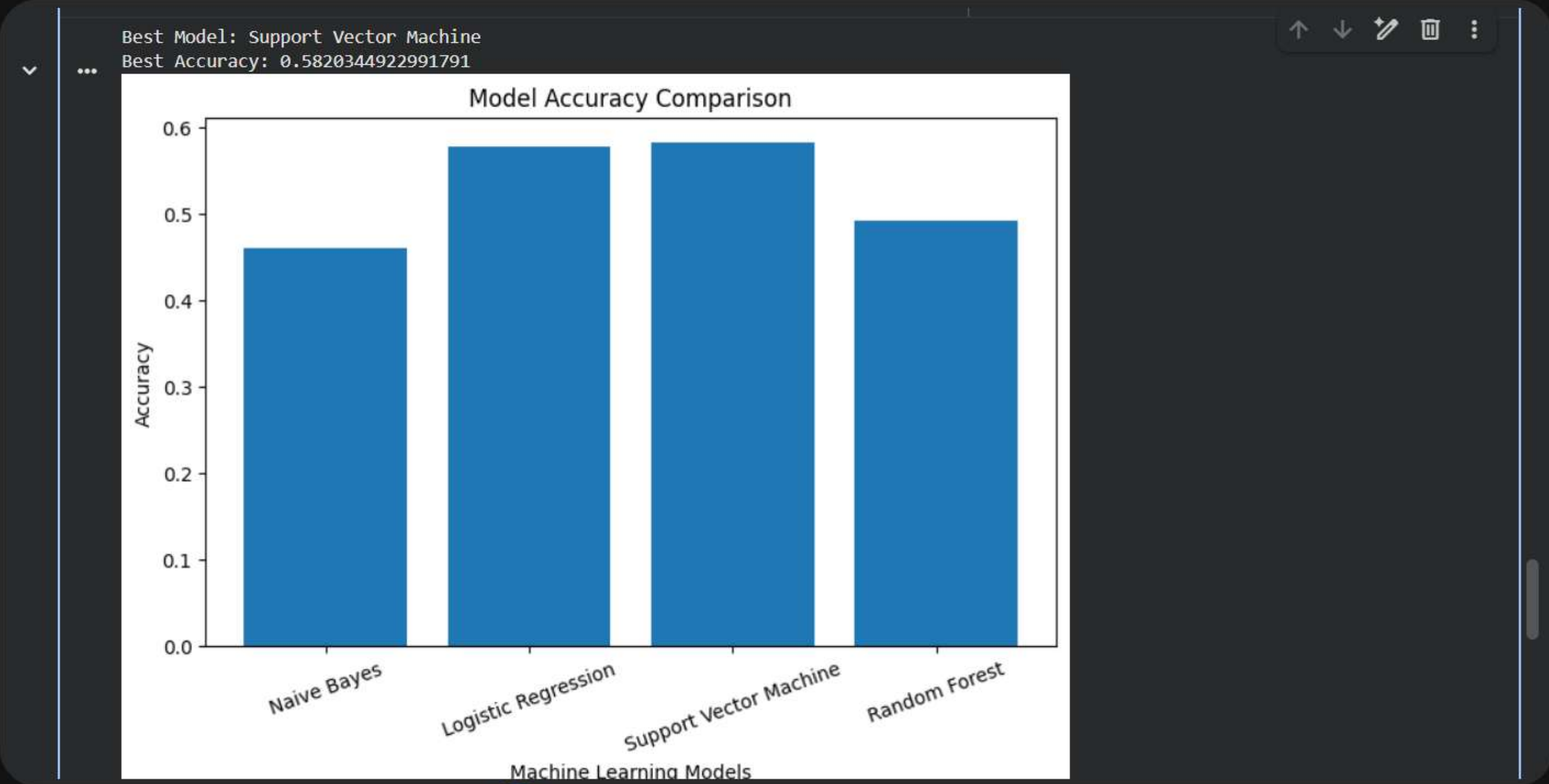
movie_genre_model.pkl

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Confusion Matrix - Best Model

0	970	7	1	0	20	0	14	6	90	1	0	0	14	0	0	0	1	4	4	9	100	8	0	4
1	152	5	0	0	200	4	230	0	0	0	3	1	0	1	0	1	0	0	7	0	0	0	0	0
2	8	833	1	0	14	0	2332	1	0	0	8	0	0	0	0	3	0	8	4	2	0	5	0	3
3	2	0	3	17	0	190	9207	2	0	0	3	0	0	0	0	0	0	8	100	0	0	0	0	0
4	0	0	0	0	3	0	3210	0	0	0	2	1	0	0	0	0	0	0	3	0	0	2	0	0
5	105	3	1	0	873	189866	1	1	1187	1	0	15	2	5613	5	6	0	3						
6	5	0	1	0	0	1451836	0	0	0	6	0	0	0	0	0	0	0	2	0	0	130	1		
7	6	3	4	1	055	18912	0	0	1	9171	0	2	122	21146	5	1	0	2						
8	243	3	3	0	2822493	4	4	0	1293	4	1	0	6104	1230	131	1	8							
9	3	0	0	1	036	02243240	0	0	3	2	1	0	0	4	1	2	8	1	2	0	0	4		
10	3	2	1	3	0	5	0	3210	1	0	0	9	0	0	0	0	0	6	100	0	1	0	0	
11	1	0	0	0	0	3	0	5	0	0	0	230	1	0	0	0	3	0	0	1	2	0	0	0
12	0	0	0	0	0	1	0348	0	0	0	2	0	1	0	0	0	2	0	0	0	0	1	0	0
13	3	1	1	0	0	30116621	0	0	0	2811	0	0	0	1	0	0	240	0	150	4				
14	0	0	0	0	0	120454	2	0	0	0	1731	0	0	2	0	0	5	1	0	0	0	0		
15	0	0	1	0	0	1008220	0	0	0	2	6	2	0	0	0	1	0	2	0	0	0	0	1	
16	2	0	2	0	0	7	2	1210	0	0	0	120	0	1	0	0	0	1	3	0	0	12	0	0
17	0	0	0	0	0	3	0183	0	0	1	0	0	1	0	0	3	0	0	3	0	4	0	0	0
18	0	0	0	0	0	2417119	0	0	0	2	0	0	0	0	431	0	112	3	0	0	0			
19	1	0	0	0	0	3003880	0	0	0	1	0	0	0	0	0	7	0	4	0	0	0	0		
20	7	0	3	1	0	1111719	0	0	0	7	1	0	1	0	1	0	479	0	0	3	0	1		
21	7	1	1	5	0	1321232522	0	0	0	171	0	0	0	4	2	63410	4	8	0	0				
22	2	0	0	0	0	4	0360	0	0	0	0	0	0	0	0	3	0	0	239	0	0	0		
23	1	0	0	0	0	7	0343	0	0	0	0	1	0	0	3	7	0	0	7	0	150	0	0	
24	12	1	3	1	0	41215381	0	0	0	300	0	2	0	1	0	3120	0	550	1					
25	3	0	0	0	0	0	07	9	0	0	0	1	0	0	0	0	0	0	2	0	0	0	4	0
26	2	0	2	0	0	5	12290	0	0	0	3	0	0	0	0	0	0	1	0	1	0	160		

Prediction results saved as prediction_results.csv

Custom Movie Genre Predictions:

Variables

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Prediction results saved as prediction_results.csv

Custom Movie Genre Predictions:

Movie Plot: A police officer fights dangerous criminals in a big city.
Predicted Genre: action

Movie Plot: A young couple falls in love but faces emotional family problems.
Predicted Genre: drama

Movie Plot: A group of friends enter a haunted house and experience terrifying events.
Predicted Genre: horror

Movie Plot: Aliens attack Earth and humans fight to save the planet.
Predicted Genre: sci-fi

Movie Plot: A poor man struggles to achieve success in life.
Predicted Genre: drama

Movie Plot: A detective investigates a mysterious murder case.
Predicted Genre: mystery

Enter a movie plot to predict genre: A poor man struggles to achieve success in life
Predicted Genre: drama

Best model saved as movie_genre_model.pkl
TF-IDF vectorizer saved as tfidf_vectorizer.pkl

===== PROJECT SUMMARY =====
Project Name: Movie Genre Classification
Dataset Used: train_data.txt
Technique Used: NLP + Machine Learning

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Best model saved as movie_genre_model.pkl
TF-IDF vectorizer saved as tfidf_vectorizer.pkl

===== PROJECT SUMMARY =====
Project Name: Movie Genre Classification
Dataset Used: train_data.txt
Technique Used: NLP + Machine Learning
Vectorizer Used: TF-IDF
Models Used:
- Naive Bayes
- Logistic Regression
- Support Vector Machine
- Random Forest
Best Model: Support Vector Machine
Best Accuracy: 0.5820344922991791
Files Generated:
- prediction_results.csv
- movie_genre_model.pkl
- tfidf_vectorizer.pkl
Task 1 Completed

Variables

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